

四. 更新过程

1. 定义: $\{X_n\}$ 独立同分布, 分布为 $F(x)$, $F(0) = P(X=0) < 1$, $0 < \mu < +\infty$

取 $T_n = \sum_{i=1}^n X_i$, $n \geq 1$, $T_0 = 0$, $N(t) = \sup\{T_n \leq t\}$ 为更新过程

性质: 1) $N(t) \geq n \Leftrightarrow T_n \leq t$

Claim: $P(N(t) < \infty) = 1$: Pf: $\mu = EX_1 \in (0, +\infty)$, $\frac{1}{n}T_n = \frac{\sum X_i}{n} \xrightarrow{a.s.} \mu$ (强大数), done

卷积公式: F 为 X 的分布函数 $\Rightarrow T_n = \sum_{i=1}^n X_i$ 的分布函数为 F 的 n 重卷积 $T_n = \sum_{i=1}^n X_i \Rightarrow T_n = t = \sum P(X_1 < t_1, \dots, X_n < t-t_1, \dots) = F_n$ (可用归纳法)

$\Rightarrow P(N(t) = n) = F_n(t) - F_{n+1}(t)$ Pf: $P(N(t) = n) = P(N(t) \geq n) - P(N(t) \geq n+1)$
 $= \int_0^t (1 - F(t-y)) dF_n(y)$ 以 T_n 为条件概率, done.

$N(t) < +\infty$ a.s. 不一定使 $EN(t) < +\infty$

期望计算: $M(t) = \sum_{n=0}^{\infty} F_n(t)$, $t \geq 0$ 收敛到 $EN(t)$

证明: 1) 若 $\{a_n\}$ 非负, 则 $\sum a_n$ 收敛或为 $+\infty$, 与求和次序无关

↓ 2) $Y = |int| \Rightarrow EY = \sum_{n=1}^{\infty} P(Y \geq n)$, easy.

$\sum F_n$ 收敛: 引理: $\forall t, \exists \alpha \in (0, 1)$, $F_n(t) = O(\alpha^n)$: Pf: $F_n(x) = P(X_1 + \dots + X_n \leq x) \leq P(X_1 \leq x, \dots, X_n \leq x) = (P(X_1 \leq x))^n$, done
 故 $\sum F_n(x) \leq \sum (F_1(x))^n = \frac{F_1(x)}{1 - F_1(x)}$ 敛于全 x

综上: $EN(t) = \sum n \cdot P(n) = \sum P(N \geq n) = \sum F_n(t)$ 敛于 $M(t)$

更新强度: $m(t) = \frac{dM(t)}{dt} = \sum f_n(t)$ ($f_n(t) = \frac{dF_n}{dt}$)

2. 更新方程

积分: $M(t) = F(t) + \int_0^t M(t-s) dF(s)$ Pf: $M(t) = \sum F_n(t) = F(t) + \sum_{n=1}^{\infty} F_n(t) = F(t) + \sum_{n=1}^{\infty} \int_0^t F_{n-1}(t-s) dF(s)$
 $m(t) = f(t) + \int_0^t m(t-s) f(s) ds$ $M(t) = \frac{dM}{dt} = f(t) + M(0) + \int_0^t m(t-s) dF(s)$ $= F(t) + \int_0^t M(t-s) dF(s)$

泛定更新方程: $K(t) = H(t) + \int_0^t K(t-s) dF(s)$, $t \geq 0$ ($K, H = 0$ 当 $t < 0$; H 在任意区间有界)

(卷积: $K(t) = H(t) + (f * K)(t)$)

唯一解: $k(t) = H(t) + \int_0^t H(t-s) dM(s)$, $M(s)$ 为 F 的 $EN(s)$

Pf: 展列: $k(t) = H(t) + (f \star k)(t) = H(t) + (f \star (H + f \star k))(t) + \dots$

$$= H(t) + (f \star H)(t) + (f \star f) \star H(t) + \dots$$

$$= H(t) + \sum_{n=1}^{\infty} f_n \star H(t) = H(t) + \int_0^t H(t-s) m(s) ds = H(t) + \int_0^t H(t-s) dM(s)$$

有界唯一性: 设另一解: $\bar{k} = H + (f \star \bar{k}) \Rightarrow$ 边界性: $\forall T > 0, |\bar{k}(t)|, |k(t)| < B, t \in [0, T] \Rightarrow \limsup_{n \rightarrow \infty} F_n(t) \leq \lim_{n \rightarrow \infty} F_n(T) = 0$

$$\text{展列: } \bar{k} = H(t) + \left(\sum_{n=1}^{\infty} f_n \star H \right)(t) + (f_n \star \bar{k})(t), \quad k - \bar{k} = \sum_{n=1}^{\infty} f_n \star (k - \bar{k})(t) \leq \int_0^t |k - \bar{k}|(t-s) dF_n(s) \leq 2B F_n(t) \rightarrow 0$$

故 $\bar{k} = k$, 唯一

Wald 恒等式: 对更新过程, $EX_1 < \infty, E(T_{N(t)+1}) = E(X_1 + \dots + X_{N(t)+1}) = EX_1 E(N(t)+1)$

$$\text{Pf: 条件期望: } E(T_{N(t)+1} | X_1 = x) = \begin{cases} x & x > t \\ x + E(T_{N(t-x)+1}) & x \leq t \end{cases}, \text{ 取 } K(t) \text{ 为 } E(T_{N(t)+1}) \Rightarrow E T_{N(t)+1} = E[E(T_{N(t)+1} | X_1)] = \int_0^{\infty} E(T_{N(t)+1} | X_1 = x) dF(x)$$

$$= \int_0^{\infty} x dF(x) + \int_0^t [x + K(t-x)] dF(x) = EX_1 + (f \star k)(t)$$

$$\text{其解为: } E(T_{N(t)+1}) = E(X_1) + \int_0^t E(X_1) dM(x) = E(X_1) \cdot (1 + M(t)) = E(X_1) E(N(t)+1)$$

余寿: $\{N(t), t \geq 0\}$ 为更新过程, $T_n = \sum_{i=1}^n X_i$ 为第 n 个 X 失效时间 $\Rightarrow r(t) = T_{N(t)+1} - t$ 为余寿命: t 时已用 $N(t)$ 个, 第 $N(t)+1$ 个 X 的余寿

$$\text{其分布为: } P(r(t) > y) = (1 - F(t+y)) + \int_0^t (1 - F(t+y-x)) dM(x)$$

$$\text{Pf: 对 } X_1 = x, \text{ 取 } R_y(t) = P(r(t) > y), \quad P(r(t) > y | X_1 = x) = \begin{cases} 1 & x \in (t+y, \infty] & (N(t)=0) \\ 0 & x \in (t, t+y] & (N(t)=0) \\ R_y(t-x) & x \in [0, t] & (N(t) \geq 1) \end{cases}$$

$$R_y(t) = E(I(r(t) > y) | X_1 = x) = \int_{t+y}^{\infty} dF(x) + \int_0^t R_y(t-x) dF(x)$$

$$\downarrow \\ R_y(t) = (1 - F(t+y)) + \int_0^t (1 - F(t+y-s)) dM(s), \text{ done.}$$

$$\text{eg. 1) } F(x) = 1 - e^{-\lambda x}, P(r(t) > y) = e^{-\lambda(t+y)} + \int_0^t e^{-\lambda(t+y-x)} \lambda dx = e^{-\lambda y}$$

2) 人口等: 略...

3. 更新定理

1. Feller 基础更新定理 $\lim_{t \rightarrow \infty} \frac{M(t)}{t} = \frac{1}{\mu}$, μ 为 EX_1 , $\mu < \infty$ 时左式为 0

$$\text{Pf: 1) 对 } \mu < \infty, T_{N(t)+1} > t \Rightarrow E(T_{N(t)+1}) = EX_1 (M(t)+1) > t \Rightarrow \frac{M(t)}{t} > \frac{1}{\mu} - \frac{1}{t} \Rightarrow \liminf_{t \rightarrow \infty} \frac{M(t)}{t} = \frac{1}{\mu}$$

$$2) \text{ 放缩: } X_n^c = \min[X_n, B] \Rightarrow X_n^c < B, X_n \Rightarrow N(t)^c = \sup\{n: T_n^c \leq t\}, T_{N(t)+1}^c = T_{N(t)}^c + X_{N(t)+1}^c \leq t + B \Rightarrow [M(t)+1] \mu_B \leq t + B$$

$$\text{控制收敛定理: } B \rightarrow \infty \text{ 时 } X_n^c \rightarrow X_n, EX_n^c \rightarrow EX_1, \text{ 且 } \mu_B \rightarrow \mu, \text{ 则 } \limsup_{t \rightarrow \infty} \frac{M(t)}{t} \leq \frac{1}{\mu} \quad \frac{M(t)}{t+B} \leq \frac{M(t)}{t} \leq \frac{1}{\mu_B} - \frac{1}{t+B}, \limsup_{t \rightarrow \infty} \frac{M(t)}{t} \leq \frac{1}{\mu_B}$$

$$\text{故 } \lim_{t \rightarrow \infty} \frac{M(t)}{t} = \frac{1}{\mu}, \text{ 若 } B \rightarrow 0 \text{ 时 } \mu_B \rightarrow \infty, \text{ 则 } \limsup_{t \rightarrow \infty} \frac{M(t)}{t} = 0 \Rightarrow \lim_{t \rightarrow \infty} \frac{M(t)}{t} = 0$$

若 $0 < E(X_1) < \infty$. $\lim_{n \rightarrow \infty} \frac{S_n}{n} = E(X_1)$ a.s., $\lim_{t \rightarrow \infty} N(t) = \infty$ a.s.

$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \frac{1}{E(X_1)}$ a.s. pf: $\frac{S_{N(t)}}{N(t)} \xrightarrow{a.s.} E(X_1)$, $t \in (S_{N(t)}, S_{N(t)+1})$, easy.

2. Blackwell(定理):

↓ 格点分布: $\exists d > 0, \sum_{n=1}^{\infty} P(X=nd) = 1$. d 为周期

1) 若不为格点分布, $\lim_{t \rightarrow \infty} M(t+a) - M(t) = \frac{a}{\mu}$ (用Feller定理)

2) 为格点分布, $\lim_{t \rightarrow \infty} P(nd \text{ 处更新}) = \frac{d}{\mu}$

3. 关键更新定理

1) $h(t) \geq 0$, 不增

$\mu = EX$, $h(t), t \geq 0$ 有: 2) $\int_0^{\infty} h(t) dt < \infty$, $H(t) = h(t) + \int_0^t H(t-s) dF(s)$

则: 1) 若不为格点分布: $\lim_{t \rightarrow \infty} H(t) = \begin{cases} \frac{1}{\mu} \int_0^{\infty} h(x) dx, & \mu < \infty \\ \infty, & \mu = \infty \end{cases}$

2) 为格点: $\lim_{t \rightarrow \infty} H(ct+nd) = \begin{cases} \frac{d}{\mu} \sum_{n=0}^{\infty} h(ct+nd), & \mu < \infty \\ \infty, & \mu = \infty \end{cases}$

pf: ? □